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Abstract

Learning equations derived from Recursive Error Minimization (REM) are applied to the identification of letters defined on a twenty-pixel field. Learning is from ten to ten thousand times faster than conventional back propagation and is guaranteed to converge to a global minimum.

A simple measure of the strength of an identification is developed in terms of output magnitude. Measures of validity (correctness) and strength are then presented as functions of training.

Introduction

In our previous paper [1], we derived REM learning equations for neural networks and applied them to simple problems. In this paper we present the results of applying REM learning to a more complex and more realistic problem.

Equations (1), (2), and (3) describe the propagation of node state forward and the propagation of error and derivative backward. Equation (4) gives the rule for correcting the value of connections. The numerator of Equation (4) is, except for the method of presenting the constants, equivalent to the "generalized delta rule." The denominator is the second derivative of the mean square error with respect to the specific connection and has a lower bound of 0.000001 to prevent computational difficulties.

$$Y_d = \tanh\left(\frac{1}{2} \sum_c Y_c w_{cd}\right) \quad (1)$$

$$E_c^* = \left(\frac{1 - Y_c^2}{2}\right) \cdot \left[E_c + \sum_d w_{cd} \cdot E_d^*\right] \quad (2)$$

$$H_c^* = \left(\frac{1 - Y_c^2}{2}\right) \cdot \left[H_c + \sum_d w_{cd} \cdot H_d^*\right] \quad (3)$$

$$\Delta w_{cd} = \frac{RM_Q \left[\frac{\beta}{P} \cdot Y_c \cdot E_d^* \right]}{RM_P \left[(Y_c \cdot H_d^*)^2 \right]} \quad (4)$$

Y_d = state of a node of the d^{th} layer

c, d = unspecified layers in a neural network

w_{cd} = connection from a node of layer c to a node of layer d

E_c^* = generalized error of a node of layer c (equivalent to the delta of the "generalized delta rule")

E_c = $D_c - Y_c$, if the node has a desired state (D_c)
 = 0, otherwise

H_c^* = generalized derivative of a node of layer c

H_c = 1, if the node has a desired state (D_c)
 = 0, otherwise

β = learning rate parameter (≈ 1.0)

RM_P [...] = recursive mean of parameter in brackets with P as parametric memory

Finally, Equation (5) incorporates the technique which guarantees convergence to a global minimum. The initial desired output is defined to be the initial output of the network, and is then smoothly changed to the final desired output as learning progresses. If the rate of change of the desired output is sufficiently small, the network is always in the neighborhood of a global minimum, and will still be in the neighborhood of a global minimum when the desired output has transitioned to the final desired output. Note that Equation (5) has been simplified from that presented previously [1].

$$D_{q,l}^n = D_{q,l}^{n-1} \cdot \left(\frac{R-1}{R} \right) + D_{q,l}^* \cdot \left(\frac{1}{R} \right) \quad (5)$$

$D_{q,l}^0$ = initial desired output (= initial observed output) of the l^{th} output node to the q^{th} experience

$D_{q,l}^*$ = final desired output of the l^{th} output node to the q^{th} experience

R = parametric memory of initial desired output

Problem

We applied REM equations to the problem of identifying the twenty-six capital letters of the English alphabet. The letter representations, shown in Figure 1, are given on a four by five grid of binary pixels, for a total of twenty pixels corresponding to twenty inputs to the network. The desired output was a number from one to twenty-six corresponding to the letters A through Z. In binary notation it requires five bits to represent twenty-six, resulting in five outputs from the network.

We used a feed-forward network of forty-five nodes: twenty input nodes, twenty intermediate nodes, and five output nodes. The intermediate and output nodes had biases modeled as the connections from a "node zero" which always had an output of one.

Results

As we pointed out previously [1], the node transfer function used here has a range from -1 to +1. This choice has no effect on the basic characteristics of the network, but does make possible a very convenient method for presentation of the validity and strength of identification. If the signs of all the output nodes are correct, the identification is valid. The product of the magnitudes of the output nodes gives a measure of strength of the identification which ranges from zero to one.

The network learned to give valid identifications in about fifteen epochs and strong identifications in thirty epochs. Figure 2 shows the distribution of identification strength with thirty epochs of training. The distribution was computed with a twenty-slot histogram, so the resolution is only 0.05 and the maximum function value is twenty. Figure 2a shows that all letters of the training set were identified with strengths greater than 0.85. In order to give some indication of the effect of imperfect input, a test set was constructed by removing each pixel from the input field, one at a time. Figure 2b presents the distribution of strength of the correct responses (91.2%) and Figure 2c presents the distribution of incorrect responses (8.8%) to this test set. Figure 3 presents the same information after ninety epochs of training. Note that the identification strengths on the training set are all greater than 0.95, but the number of correct responses to the test set is somewhat lower (89.2%).

It is obvious that if the training set is redefined to include the test set, correct responses should be obtained for all cases. Figure 4 shows that thirty epochs of training on this expanded set gives valid identifications with strengths greater than 0.85 for the original training set and strengths greater than 0.65 for the set with missing pixels.

Finally, we made a comparison with conventional back propagation using the "bp" program supplied by McClelland and Rumelhart [2] with the identical network configuration on the same problem. Figure 5 presents the epochs of training vs. RMS error for REM learning and conventional back propagation. The conventional back propagation is extrapolated from 7500 epochs (twenty-one hours) to 900,000 epochs. Note that REM is exponentially convergent and back propagation is power convergent. Thus the ratio of epochs required varies from 0.1 to 0.0001, depending on the allowable error.

We estimate the computational load per epoch for the REM equations to be about twice that of the back propagation equations. This is based on observed running time on an IBM AT. The back propagation program, written in C and optimized for speed, runs at ten seconds per epoch for this problem. The REM program, written in Pascal with no consideration given to speed, runs at forty seconds per epoch, with an assumed factor of two improvement with optimization.

Conclusion

Learning with REM equations has been demonstrated to be orders of magnitude faster than learning with conventional back propagation equations, both in number of epochs required and total computational time. The advantage of the REM equations increases at lower levels of acceptable error. The node transfer equation chosen for this work is shown to provide a convenient method of presenting the validity and strength of identification.

References

- [1] Simon, W., and Carter, J., "Back Propagation Learning Equations from the Minimization of Recursive Error," *Proceedings of the IEEE International Conference on Systems Engineering*, August 24-26, 1989.
- [2] McClelland, J., and Rumelhart, D., *Explorations in Parallel Distributed Processing*, MIT Press, 1988.

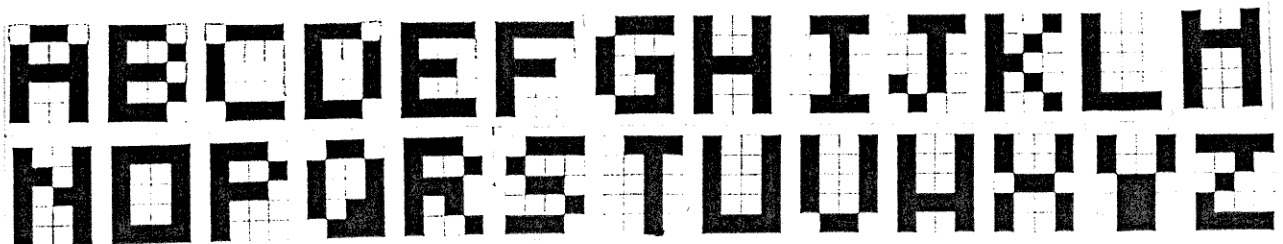
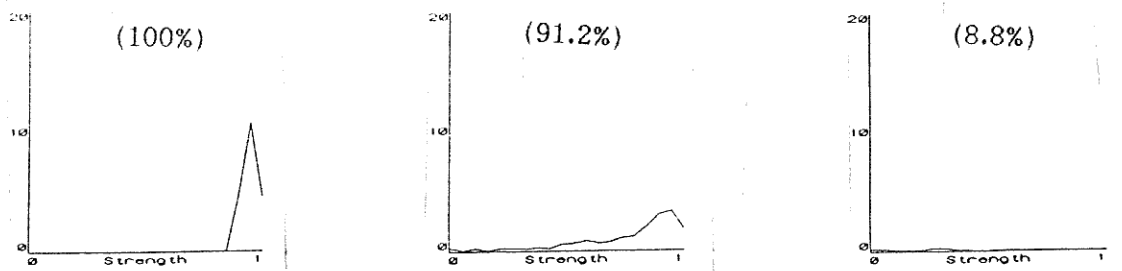
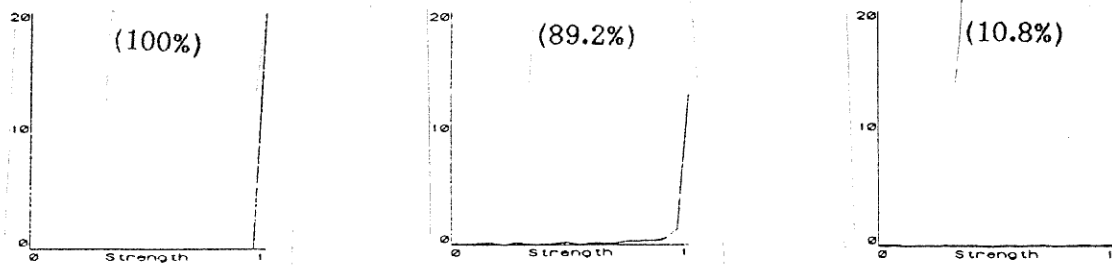


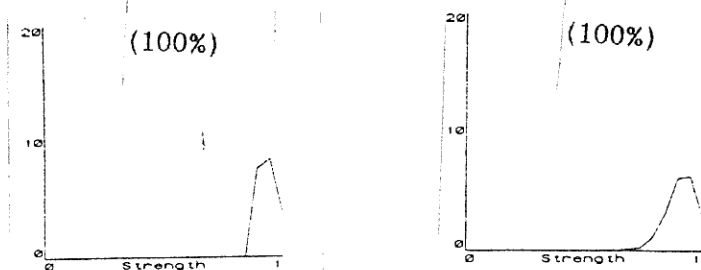
Figure 1. Capital Letters Given on a Four by Five Pixel Grid



a) Training Set--Valid b) Test Set--Valid c) Test Set--Invalid
Figure 2. Identification Strength--Thirty Epochs of Training Set



a) Training Set--Valid b) Test Set--Valid c) Test Set--Invalid
Figure 3. Identification Strength--Ninety Epochs of Training Set



a) Training Set--Valid b) Test Set--Valid
Figure 4. Identification Strength--Thirty Epochs of Training and Test Sets

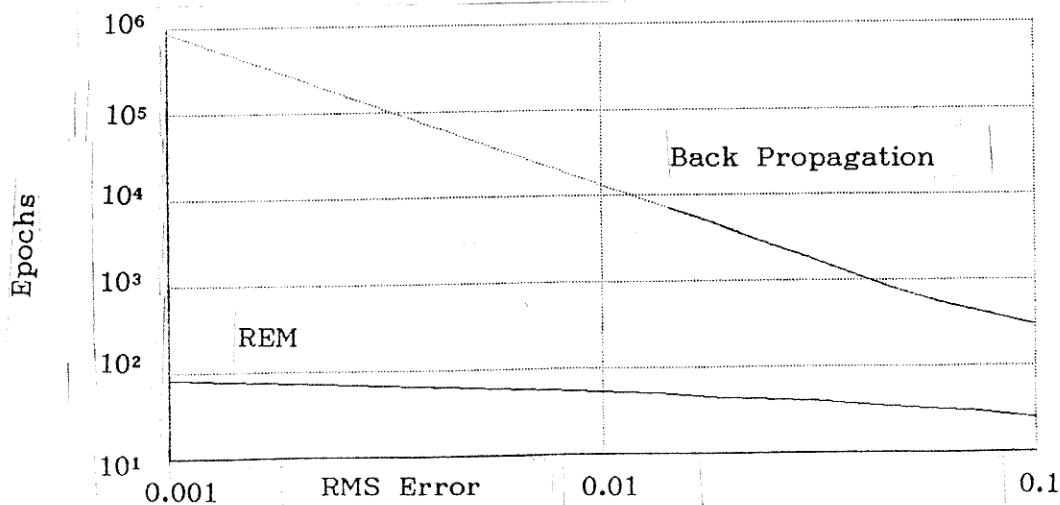


Figure 5. Required Training as a Function of Allowable RMS Error